Internship Title: 2025

Internship Program Name: Cyber Security

Your Name: Areesh Awan

Internship Lead Name: Faizyab



info@emrchains.com

office#1401, wing#04, NSTP, NUST,
H.12, Islamabad, Pakistan

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info@emrchains.com

office#1401, wing#04, NSTP, NUST,
H.12, Islamabad, Pakistan

## 1. Introduction

In today's digital world, network security is a critical aspect of cybersecurity. Nmap (Network Mapper) is one of the most powerful and widely used open-source tools for network scanning, host discovery, and security auditing. By watching the Nmap video tutorial, I gained a practical understanding of how this tool works and how it is used by cybersecurity professionals to detect vulnerabilities and manage network security.

## 2. Task details and working


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```
┌──(kali㉿kali)-[~]
└─$ sudo apt install libssl-dev autoconf make g++ subversion
make is already the newest version (4.4.1-2).
make set to manually installed.
g++ is already the newest version (4:14.2.0-1).
g++ set to manually installed.
subversion is already the newest version (1.14.5-3).
subversion set to manually installed.
The following packages were automatically installed and are no longer required:
  crackmapexec            libflac12t64                libgtksourceviewmm-3.0-0v5  libpoppler140           libwebrtc-audio-processing1        python3-setproctitle
  firebird3.0-common      libfmt9                     libgumbo2                   libpython3.12-dev       libx265-209                        python3-tomlkit
  firebird3.0-common-doc  libfuse3-3                  libhdf5-103-1t64            libpython3.12-minimal   openjdk-23-jre                     python3-wheel-whl
  icu-devtools            libgdal35                   libhdf5-hl-100t64           libpython3.12-stdlib    openjdk-23-jre-headless            python3.12
  libabsl20230802         libgeos3.13.0               libicu-dev                  libpython3.12t64        python3-aioconsole                 python3.12-dev
  libbfio1                libgl1-mesa-dev             libjxl0.9                   libqt5sensors5          python3-appdirs                    python3.12-minimal
  libc++1-19              libglapi-mesa               liblbfgsb0                  libqt5webkit5           python3-dunamai                    python3.12-tk
  libc++abi1-19           libgles-dev                 libldap-2.5-0               libsuperlu6             python3-nfsclient                  python3.12-venv
  libcapstone4            libgles1                    libmbedcrypto7t64           libtag1v5               python3-ntlm-auth                  ruby-zeitwerk
  libconfig++9v5          libglvnd-core-dev           libmsgraph-0-1              libtag1v5-vanilla       python3-packaging-whl              ruby3.1
  libconfig9              libglvnd-dev                libnetcdf19t64              libtagc0                python3-poetry-dynamic-versioning  ruby3.1-dev
  libdirectfb-1.7-7t64    libgtksourceview-3.0-1      libopenh264-7               libunwind-19            python3-pywerview                  ruby3.1-doc
  libegl-dev              libgtksourceview-3.0-common libpaper1                   libutempter0            python3-requests-ntlm              strongswan
Use 'sudo apt autoremove' to remove them.

Installing:
  autoconf  libssl-dev

Installing dependencies:
  automake  autotools-dev
```

```
┌──(kali㉿kali)-[~]
└─$ svn co https://svn.nmap.org/nmap/
A    nmap/timing.cc
A    nmap/nmap-services
A    nmap/MACLookup.cc
A    nmap/payload.h
A    nmap/nse_db.h
A    nmap/libpcre
A    nmap/libpcre/install-sh
A    nmap/libpcre/config-cmake.h.in
A    nmap/libpcre/m4
A    nmap/libpcre/m4/ltoptions.m4
A    nmap/libpcre/m4/ax_pthread.m4
A    nmap/libpcre/m4/ltsugar.m4
A    nmap/libpcre/m4/pcre2_visibility.m4
A    nmap/libpcre/m4/libtool.m4
A    nmap/libpcre/m4/ltversion.m4
A    nmap/libpcre/m4/lt~obsolete.m4
A    nmap/libpcre/cmake
A    nmap/libpcre/cmake/COPYING-CMAKE-SCRIPTS
A    nmap/libpcre/cmake/pcre2-config.cmake.in
A    nmap/libpcre/cmake/pcre2-config-version.cmake.in
A    nmap/libpcre/MODULE.bazel
A    nmap/libpcre/configure
A    nmap/libpcre/Makefile.in
A    nmap/libpcre/pcre2-config.in
A    nmap/libpcre/AUTHORS.md
A    nmap/libpcre/build.zig
A    nmap/libpcre/depcomp
A    nmap/libpcre/src
A    nmap/libpcre/src/pcre2_maketables.c
A    nmap/libpcre/src/pcre2_find_bracket.c
A    nmap/libpcre/src/pcre2_substitute.c
A    nmap/libpcre/src/pcre2_ucd.c
```

```
A    nmap/docs/licenses
A    nmap/docs/licenses/PCRE-license.txt
A    nmap/docs/licenses/MIT
A    nmap/docs/licenses/BSD-simplified
A    nmap/docs/licenses/LGPL-2.1
A    nmap/docs/licenses/LGPL-2
A    nmap/docs/licenses/zlib-license.txt
A    nmap/docs/licenses/LIBLINEAR-license.txt
A    nmap/docs/licenses/Libdnet-license.txt
A    nmap/docs/licenses/Lua-license.txt
A    nmap/docs/licenses/WinPcap-license.txt
A    nmap/docs/licenses/OpenSSL-license.txt
A    nmap/docs/licenses/BSD-modified.txt
A    nmap/docs/licenses/MPL-1.1
A    nmap/docs/3rd-party-licenses.txt
A    nmap/docs/leet-nmap-ascii-art-pc.txt
A    nmap/docs/leet-nmap-ascii-art.txt
A    nmap/docs/nmap.usage.txt
A    nmap/docs/win32-installer-zenmap-buildguide.txt
A    nmap/docs/device-types.txt
A    nmap/config.guess
A    nmap/README-WIN32
A    nmap/charpool.h
A    nmap/targets.cc
A    nmap/scan_engine.h
A    nmap/nse_fs.cc
A    nmap/idle_scan.cc
A    nmap/nse_ssl_cert.cc
A    nmap/lpeg.c
 U   nmap
Checked out revision 39255.
```

```
┌──(kali㉿kali)-[~]
└─$ cd nmap

┌──(kali㉿kali)-[~/nmap]
└─$ ls
acinclude.m4   HACKING           missing                NmapOps.h          nse_fs.h         nse_zlib.cc      scan_engine.cc          Target.h
aclocal.m4     idle_scan.cc      mswin32                nmap-os-db         nselib           nse_zlib.h       scan_engine_connect.cc  targets.cc
BSDmakefile    idle_scan.h       nbase                  NmapOutputTable.cc nsock            osscan2.cc       scan_engine_connect.h   targets.h
CHANGELOG      INSTALL           ncat                   NmapOutputTable.h  nse_libssh2.cc   osscan2.h        scan_engine.h           tcpip.cc
charpool.cc    libdnet-stripped  ndiff                  nmap-protocols     nse_libssh2.h    osscan.cc        scan_engine_raw.cc      tcpip.h
charpool.h     liblinear         NewTargets.cc          nmap-rpc           nse_lpeg.cc      osscan.h         scan_engine_raw.h       tests
checklibs.sh   liblua            NewTargets.h           nmap-service-probes nse_lpeg.h      output.cc        scan_lists.cc           timing.cc
config.guess   libnetutil        nmap_amigaos.h         nmap-services      nse_lua.h        output.h         scan_lists.h            timing.h
config.sub     libpcap           nmap.cc                nmap.spec.in       nse_main.cc      payload.cc       scripts                 todo
configure      libpcre           nmap_config.h.in       nmap_tty.cc        nse_main.h       payload.h        service_scan.cc         traceroute.cc
configure.ac   libssh2           nmap_dns.cc            nmap_tty.h         nse_main.lua     portlist.cc      service_scan.h          traceroute.h
CONTRIBUTING.md libz             nmap_dns.h             nmap_winconfig.h   nse_nmaplib.cc   portlist.h       services.cc             utils.cc
depcomp        LICENSE           nmap_error.cc          nping              nse_nmaplib.h    portreasons.cc   services.h              utils.h
docs           lpeg.c            nmap_error.h           nse_db.cc          nse_nsock.cc     portreasons.h    shtool                  xml.cc
FingerPrintResults.cc ltmain.sh  nmap_ftp.cc            nse_db.h           nse_nsock.h      probespec.h      string_pool.cc          xml.h
FingerPrintResults.h MACLookup.cc nmap_ftp.h            nse_debug.cc       nse_openssl.cc   protocols.cc     string_pool.h           zenmap
FPEngine.cc    MACLookup.h       nmap.h                 nse_debug.h        nse_openssl.h    protocols.h      struct_ip.h             zenmap.spec.in
FPEngine.h     macosx            nmap-header-template.cc nse_dnet.cc       nse_ssl_cert.cc  protocols.h      Target.cc
FPModel.cc     main.cc           nmap-mac-prefixes      nse_dnet.h         nse_ssl_cert.h   README.md        TargetGroup.cc
FPModel.h      Makefile.in       NmapOps.cc             nse_fs.cc          nse_utility.cc   README-WIN32     TargetGroup.h
                                                                           nse_utility.h
```

```
┌──(kali㉿kali)-[~/nmap]
└─$ ./configure --help
`configure' configures this package to adapt to many kinds of systems.

Usage: ./configure [OPTION]... [VAR=VALUE]...

To assign environment variables (e.g., CC, CFLAGS...), specify them as
VAR=VALUE.  See below for descriptions of some of the useful variables.

Defaults for the options are specified in brackets.

Configuration:
  -h, --help              display this help and exit
      --help=short        display options specific to this package
      --help=recursive    display the short help of all the included packages
  -V, --version           display version information and exit
  -q, --quiet, --silent   do not print `checking ...' messages
      --cache-file=FILE   cache test results in FILE [disabled]
  -C, --config-cache      alias for `--cache-file=config.cache'
  -n, --no-create         do not create output files
      --srcdir=DIR        find the sources in DIR [configure dir or `..']

Installation directories:
  --prefix=PREFIX         install architecture-independent files in PREFIX
                          [/usr/local]
  --exec-prefix=EPREFIX   install architecture-dependent files in EPREFIX
                          [PREFIX]

By default, `make install' will install all the files in
`/usr/local/bin', `/usr/local/lib' etc.  You can specify
an installation prefix other than `/usr/local' using `--prefix',
for instance `--prefix=$HOME'.

For better control, use the options below.
```

---

```
  --with-libdnet=included Use the libdnet version included with Nmap (default)
  --with-liblua=DIR       Use an existing (compiled) lua lib from DIR/include
                          and DIR/lib.
  --with-liblua=included  Use the liblua version included with Nmap
  --without-liblua        Compile without lua (this will exclude all of NSE
                          from compilation)
  --with-liblinear=DIR    Use an existing (compiled) liblinear from
                          DIR/include and DIR/lib.
  --with-liblinear=included
                          Use the liblinear version included with Nmap
  --with-libnbase=DIR     Look for nbase include/libs in DIR
  --with-libnsock=DIR     Compile and link to libnsock in DIR
  --without-ncat          Skip build and installation of Ncat

Some influential environment variables:
  CC          C compiler command
  CFLAGS      C compiler flags
  LDFLAGS     linker flags, e.g. -L<lib dir> if you have libraries in a
              nonstandard directory <lib dir>
  LIBS        libraries to pass to the linker, e.g. -l<library>
  CPPFLAGS    (Objective) C/C++ preprocessor flags, e.g. -I<include dir> if
              you have headers in a nonstandard directory <include dir>
  CXX         C++ compiler command
  CXXFLAGS    C++ compiler flags
  PYTHON      the Python interpreter
  CPP         C preprocessor

Use these variables to override the choices made by `configure' or to help
it to find libraries and programs with nonstandard names/locations.

Report bugs to the package provider.
```

```
┌──(kali㉿kali)-[~/nmap]
└─$ ./configure
checking whether NLS is requested ... yes
checking build system type ... x86_64-unknown-linux-gnu
checking host system type ... x86_64-unknown-linux-gnu
checking for gcc ... gcc
checking whether the C compiler works ... yes
checking for C compiler default output file name ... a.out
checking for suffix of executables ...
checking whether we are cross compiling ... no
checking for suffix of object files ... o
checking whether the compiler supports GNU C ... yes
checking whether gcc accepts -g ... yes
checking for gcc option to enable C11 features ... none needed
checking for inline ... inline
checking for gcc ... (cached) gcc
checking whether the compiler supports GNU C ... (cached) yes
checking whether gcc accepts -g ... (cached) yes
checking for gcc option to enable C11 features ... (cached) none needed
checking for g++ ... g++
checking whether the compiler supports GNU C++ ... yes
checking whether g++ accepts -g ... yes
checking for g++ option to enable C++11 features ... none needed
checking for ranlib ... ranlib
checking for a BSD-compatible install ... /usr/bin/install -c
checking for gawk ... gawk
checking for __func__ ... yes
checking for strip ... /usr/bin/strip
checking for stdio.h ... yes
checking for stdlib.h ... yes
checking for string.h ... yes
checking for inttypes.h ... yes
checking for stdint.h ... yes
```

```
                    \`-"'"-'/
                     } 6 6 {
                   =. Y ,=
                     /^^^\  .
                    /     \  )  Ncat: A modern interpretation of classic Netcat
                   (  )-(  )/
                    -""___""___/
                     /   Ncat    \_/
                   (
                    \_.=|____E
Configuration complete.
                        __.-------.__
                     _.-'_.--;--.__ `-._
                  .-'_.-'  /  .+.  \ `-._ `-.
                .'.-'  -  |--|o--|  `-. `.
               (_ <O__  \  `+'  /      __O> _)
                 `--._ `--..__`.|.'__..-'' _.--'
                      ``--.__ ____ .--''
 ____ ____ ____  ____       _       _______
|_  /|_  _||_  _|/    _|     / \     |_   __ \
  | |  \ | |  \ |  V    |     / _ \      | |__) |
  | |\ \| |  | |\  /| |     / ___ \     |  ___/
 _| |_\   |_ _| |_\/_| |_  _/ /   \ \_  _| |_
|_____|\____||_____||_____||____| |____||_____|

 NMAP IS A POWERFUL TOOL -- USE CAREFULLY AND RESPONSIBLY
Configured with: ndiff zenmap nping openssl zlib libssh2 lua ncat
Configured without: localdirs
Type make (or gmake on some *BSD machines) to compile.

┌──(kali㉿kali)-[~/nmap]
└─$ make
cd . && autoconf
configure.ac:15: warning: The macro 'AC_HELP_STRING' is obsolete.
configure.ac:15: You should run autoupdate.
./lib/autoconf/general.m4:204: AC_HELP_STRING is expanded from...
configure.ac:15: the top level
configure.ac:58: warning: The macro 'AC_CONFIG_HEADER' is obsolete.
configure.ac:58: You should run autoupdate.
./lib/autoconf/status.m4:719: AC_CONFIG_HEADER is expanded from...
configure.ac:58: the top level
configure.ac:85: warning: The macro 'AC_TRY_COMPILE' is obsolete.
configure.ac:85: You should run autoupdate.
./lib/autoconf/general.m4:2845: AC_TRY_COMPILE is expanded from...
configure.ac:85: the top level
configure.ac:94: warning: The macro 'AC_TRY_COMPILE' is obsolete.
configure.ac:94: You should run autoupdate.
./lib/autoconf/general.m4:2845: AC_TRY_COMPILE is expanded from...
configure.ac:94: the top level
configure.ac:250: warning: The macro 'AC_HELP_STRING' is obsolete.
configure.ac:250: You should run autoupdate.
./lib/autoconf/general.m4:204: AC_HELP_STRING is expanded from...
configure.ac:250: the top level
configure.ac:288: warning: The macro 'AC_HELP_STRING' is obsolete.
configure.ac:288: You should run autoupdate.
./lib/autoconf/general.m4:204: AC_HELP_STRING is expanded from...
configure.ac:288: the top level
configure.ac:327: warning: The macro 'AC_HELP_STRING' is obsolete.
configure.ac:327: You should run autoupdate.
./lib/autoconf/general.m4:204: AC_HELP_STRING is expanded from...
configure.ac:327: the top level
configure.ac:359: warning: The macro 'AC_HELP_STRING' is obsolete.
configure.ac:359: You should run autoupdate.
```

```
┌──(kali㉿kali)-[~/nmap]
└─$ sudo apt install pipx
pipx install setuptools==62.0.0

pipx is already the newest version (1.7.1-1).
pipx set to manually installed.
The following packages were automatically installed and are no longer required:
  crackmapexec   libflac12t64   libgtksourceviewmm-3.0-0v5  libpoppler140   libwebrtc-audio-processing1   python3-setproctitle
  firebird3.0-common   libfmt9   libgumbo2   libpython3.12-dev   libx265-209   python3-tomlkit
  firebird3.0-common-doc   libfuse3-3   libhdf5-103-1t64   libpython3.12-minimal   openjdk-23-jre   python3-wheel-whl
  icu-devtools   libgdal35   libhdf5-hl-100t64   libpython3.12-stdlib   openjdk-23-jre-headless   python3.12
  libabsl20230802   libgeos3.13.0   libicu-dev   libpython3.12t64   python3-aioconsole   python3.12-dev
  libbfio1   libgl1-mesa-dev   libjxl0.9   libqt5sensors5   python3-appdirs   python3.12-minimal
  libc++1-19   libglapi-mesa   liblbfgsb0   libqt5webkit5   python3-dunamai   python3.12-tk
  libc++abi1-19   libgles-dev   libldap-2.5-0   libsuperlu6   python3-nfsclient   python3.12-venv
  libcapstone4   libgles1   libmbedcrypto7t64   libtag1v5   python3-ntlm-auth   ruby-zeitwerk
  libconfig++9v5   libglvnd-core-dev   libmsgraph-0-1   libtag1v5-vanilla   python3-packaging-whl   ruby3.1
  libconfig9   libglvnd-dev   libnetcdf19t64   libtagc0   python3-poetry-dynamic-versioning   ruby3.1-dev
  libdirectfb-1.7-7t64   libgtksourceview-3.0-1   libopenh264-7   libunwind-19   python3-pywerview   ruby3.1-doc
  libegl-dev   libgtksourceview-3.0-common   libpaper1   libutempter0   python3-requests-ntlm   strongswan
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 166

No apps associated with package setuptools or its dependencies. If you are attempting to install a library, pipx should not be used. Consider using pip or a similar
tool instead.
```

```
┌──(kali㉿kali)-[~/nmap]
└─$ sudo make install
Compiling libnetutil
cd libnetutil && make
make[1]: Entering directory '/home/kali/nmap/libnetutil'
make[1]: Nothing to be done for 'all'.
make[1]: Leaving directory '/home/kali/nmap/libnetutil'
Compiling liblinear
make[1]: Entering directory '/home/kali/nmap/liblinear'
make[1]: 'liblinear.a' is up to date.
make[1]: Leaving directory '/home/kali/nmap/liblinear'
Compiling libpcap
make[1]: Entering directory '/home/kali/nmap/libpcap'
VER=`cat ./VERSION`; \
MAJOR_VER=`sed 's/\([0-9][0-9]*\)\..*/\1/' ./VERSION`; \
gcc  -shared -Wl,-soname,libpcap.so.$MAJOR_VER \
    -o libpcap.so.$VER pcap-linux.o fad-getad.o pcap-usb-linux.o pcap-netfilter-linux.o  pcap.o gencode.o optimize.o nametoaddr.o etherent.o fmtutils.o pcap-util.o sa
vefile.o sf-pcap.o sf-pcapng.o pcap-common.o bpf_image.o bpf_filter.o bpf_dump.o scanner.o grammar.o
cat: ./VERSION: No such file or directory
sed: can't read ./VERSION: No such file or directory
make[1]: Leaving directory '/home/kali/nmap/libpcap'
Compiling libssh2
make[1]: Entering directory '/home/kali/nmap/libssh2/src'
make  all-am
make[2]: Entering directory '/home/kali/nmap/libssh2/src'
make[2]: Leaving directory '/home/kali/nmap/libssh2/src'
make[1]: Leaving directory '/home/kali/nmap/libssh2/src'
cd libssh2/src && make libdir="/lib" prefix="" DESTDIR=/home/kali/nmap/libssh2 install-libLTLIBRARIES;
make[1]: Entering directory '/home/kali/nmap/libssh2/src'
 /usr/bin/mkdir -p '/home/kali/nmap/libssh2/lib'
 /bin/bash ../libtool   --mode=install /usr/bin/install -c   libssh2.la '/home/kali/nmap/libssh2/lib'
libtool: install: /usr/bin/install -c .libs/libssh2.so.1.0.1 /home/kali/nmap/libssh2/lib/libssh2.so.1.0.1
libtool: install: (cd /home/kali/nmap/libssh2/lib && { ln -s -f libssh2.so.1.0.1 libssh2.so.1 || { rm -f libssh2.so.1 && ln -s libssh2.so.1.0.1 libssh2.so.1; }; })
```

```
┌──(kali㉿kali)-[~/nmap]
└─$ cd zenmap

┌──(kali㉿kali)-[~/nmap/zenmap]
└─$ ls
COPYING_HIGWIDGETS  install_scripts  LICENSE  pyproject.toml  radialnet  README  test  zenmap  zenmapCore  zenmapGUI

┌──(kali㉿kali)-[~/nmap/zenmap]
└─$ ./zenmap
/home/kali/nmap/zenmap/zenmapCore/I18N.py:77: DeprecationWarning: 'locale.getdefaultlocale' is deprecated and slated for removal in Python 3.15. Use setlocale(), g
encoding() and getlocale() instead.
  loc, enc = locale.getdefaultlocale()
/home/kali/nmap/zenmap/zenmapGUI/Icons.py:117: PyGTKDeprecationWarning: Gtk.IconSet(pixbuf) has been deprecated. Please use: Gtk.IconSet.new_from_pixbuf(pixbuf)
  iconset = Gtk.IconSet(pixbuf=pixbuf)
/usr/lib/python3/dist-packages/gi/overrides/Gtk.py:1591: DeprecationWarning: Gtk.IconSet.new_from_pixbuf is deprecated
  iconset = Gtk.IconSet.new_from_pixbuf(pixbuf)
/home/kali/nmap/zenmap/zenmapGUI/Icons.py:118: DeprecationWarning: Gtk.IconFactory.add is deprecated
  iconfactory.add(key, iconset)
/home/kali/nmap/zenmap/zenmapGUI/Icons.py:120: DeprecationWarning: Gtk.IconFactory.add_default is deprecated
  iconfactory.add_default()
/home/kali/nmap/zenmap/zenmapGUI/ScanToolbar.py:125: DeprecationWarning: Gtk.Container.set_focus_chain is deprecated
  self.target_entry.set_focus_chain((self.target_entry.get_child(),))
/home/kali/nmap/zenmap/zenmapGUI/ScanOpenPortsPage.py:135: DeprecationWarning: Gtk.ScrolledWindow.add_with_viewport is deprecated
  self.add_with_viewport(self.host)
/home/kali/nmap/zenmap/radialnet/bestwidgets/labels.py:89: DeprecationWarning: Gtk.Misc.set_alignment is deprecated
  self.set_alignment(0, 0.50)
/home/kali/nmap/zenmap/radialnet/bestwidgets/expanders.py:77: DeprecationWarning: Gtk.Alignment.new is deprecated
  self.__alignment = Gtk.Alignment.new(0, 0, 1, 1)
/home/kali/nmap/zenmap/radialnet/bestwidgets/expanders.py:78: DeprecationWarning: Gtk.Alignment.set_padding is deprecated
  self.__alignment.set_padding(12, 0, 24, 0)
/home/kali/nmap/zenmap/radialnet/bestwidgets/boxes.py:150: DeprecationWarning: Gtk.Table.set_row_spacings is deprecated
  self.set_row_spacings(spacing)
/home/kali/nmap/zenmap/radialnet/bestwidgets/boxes.py:151: DeprecationWarning: Gtk.Table.set_col_spacings is deprecated
```

```
NMAP(1)                                Nmap Reference Guide                                NMAP(1)

NAME
       nmap - Network exploration tool and security / port scanner

SYNOPSIS

       nmap [Scan Type ... ] [Options] {target specification}

DESCRIPTION
       Nmap ("Network Mapper") is an open source tool for network exploration and security auditing. It was designed to rapidly scan large networks, although it
       works fine against single hosts. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name
       and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens
       of other characteristics. While Nmap is commonly used for security audits, many systems and network administrators find it useful for routine tasks such as
       network inventory, managing service upgrade schedules, and monitoring host or service uptime.

       The output from Nmap is a list of scanned targets, with supplemental information on each depending on the options used. Key among that information is the
       "interesting ports table".  That table lists the port number and protocol, service name, and state. The state is either open, filtered, closed, or unfiltered.
       Open means that an application on the target machine is listening for connections/packets on that port.  Filtered means that a firewall, filter, or other
       network obstacle is blocking the port so that Nmap cannot tell whether it is open or closed.  Closed ports have no application listening on them, though they
       could open up at any time. Ports are classified as unfiltered when they are responsive to Nmap's probes, but Nmap cannot determine whether they are open or
       closed. Nmap reports the state combinations open|filtered and closed|filtered when it cannot determine which of the two states describe a port. The port table
       may also include software version details when version detection has been requested. When an IP protocol scan is requested (-sO), Nmap provides information on
       supported IP protocols rather than listening ports.

       In addition to the interesting ports table, Nmap can provide further information on targets, including reverse DNS names, operating system guesses, device
       types, and MAC addresses.

       A typical Nmap scan is shown in Example 1. The only Nmap arguments used in this example are -A, to enable OS and version detection, script scanning, and
       traceroute; -T4 for faster execution; and then the hostname.

       Example 1. A representative Nmap scan

 Manual page nmap(1) line 1 (press h for help or q to quit)
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -P 22 80 192.168.100.58
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:03 EDT
Nmap scan report for 192.168.100.58
Host is up (0.0000030s latency).
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 3 IP addresses (1 host up) scanned in 6.36 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.100.58
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:05 EDT
Nmap scan report for 192.168.100.58
Host is up (0.000011s latency).
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -O tesla.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:07 EDT
Stats: 0:00:19 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 58.05% done; ETC: 15:07 (0:00:13 remaining)
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 80.90% done; ETC: 15:07 (0:00:05 remaining)
Nmap scan report for tesla.com (2.18.53.207)
Host is up (0.50s latency).
Other addresses for tesla.com (not scanned): 2.18.52.207 2.18.55.207 2.18.51.207 2.18.50.207 2.18.48.207 2.18.49.207 2.18.54.207 23.40.100.207 23.7.244.207
rDNS record for 2.18.53.207: a2-18-53-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 4.19 - 5.15 (95%), Linux 5.0 - 5.14 (91%), Linux 4.15 - 5.19 (91%), Linux 4.15 (91%), IPFire 2.27 (Linux 5.15 - 6.1) (89%), Linux 5.4 (89%
), Dish Network Hopper media device (87%), Linux 2.6.32 (87%), Linux 2.6.32 - 2.6.35 (87%), Linux 2.6.39 (87%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 21 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 35.78 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS geotv.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:11 EDT
Nmap scan report for geotv.com (185.53.178.50)
Host is up (0.16s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https

Nmap done: 1 IP address (1 host up) scanned in 14.64 seconds

┌──(kali㉿kali)-[~]
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sC 192.168.100.58
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:12 EDT
Nmap scan report for 192.168.100.58
Host is up (0.0000090s latency).
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -A vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:14 EDT
Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.85% done; ETC: 15:34 (0:19:27 remaining)
Stats: 0:02:51 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.70% done; ETC: 15:56 (0:39:41 remaining)
Stats: 0:02:53 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.75% done; ETC: 15:56 (0:39:36 remaining)
Stats: 0:02:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.85% done; ETC: 15:57 (0:39:40 remaining)
Stats: 0:02:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.90% done; ETC: 15:56 (0:39:35 remaining)
Stats: 0:02:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.95% done; ETC: 15:56 (0:39:30 remaining)
Stats: 0:03:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 7.05% done; ETC: 15:56 (0:39:33 remaining)
Stats: 0:03:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 7.10% done; ETC: 15:57 (0:39:41 remaining)
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -f tesla.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-20 15:22 EDT
Nmap scan report for tesla.com (2.18.51.207)
Host is up (0.27s latency).
Other addresses for tesla.com (not scanned): 2.18.50.207 2.18.55.207 2.18.48.207 2.18.53.207 23.7.244.207 2.18.49.207 2.18.52.207 23.40.100.207 2.18.54.207
rDNS record for 2.18.51.207: a2-18-51-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 18.49 seconds
```

## Scan a list of target

```
┌──(root㉿kali)-[/home/kali]
└─# nano ip.txt

┌──(root㉿kali)-[/home/kali]
└─# cat ip.txt
192.168.100.58
192.168.100.59

┌──(root㉿kali)-[/home/kali]
└─# nmap -F -iL ip.txt
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 12:54 -0400
Nmap scan report for 192.168.100.58
Host is up.
All 100 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)

Nmap done: 2 IP addresses (1 host up) scanned in 27.44 seconds
```

## Exclude target from scan

```
┌──(root㉿kali)-[/home/kali]
└─# nmap 192.168.100.59/24 --exclude 192.168.100.58

Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 13:01 -0400
Nmap done: 255 IP addresses (0 hosts up) scanned in 12.44 seconds
```

## Selecting a Interface Network

```
┌──(root㉿kali)-[/home/kali]
└─# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 7e:e8:9d:e2:a1:4d brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.58/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
       valid_lft 69715sec preferred_lft 69715sec
    inet6 fe80::51e6:133:fb12:a4cd/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

## Scan ipv6

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -6 fe80::51e6:133:fb12:a4cd
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 14:11 -0400
Nmap scan report for fe80::51e6:133:fb12:a4cd
Host is up (0.000031s latency).
All 1000 scanned ports on fe80::51e6:133:fb12:a4cd are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 4.80 seconds
```

## Scan Random Target

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -iR 3
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 14:14 -0400
Nmap done: 3 IP addresses (0 hosts up) scanned in 5.38 seconds
```

## Display Port State Random Ports

```
┌──(root㉿kali)-[/home/kali]
└─# nmap --reason 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 14:17 -0400
Stats: 0:01:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 34.50% done; ETC: 14:21 (0:02:15 remaining)
Nmap scan report for 192.168.100.58
Host is up, received localhost-response.
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 208.04 seconds
```

## Only Display Open Ports

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -F --open 192.168.100.58

Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-22 14:48 -0400
Stats: 0:00:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 60.00% done; ETC: 14:48 (0:00:09 remaining)
Nmap done: 1 IP address (1 host up) scanned in 26.12 seconds
```

## For Top Ports

```
┌──(root㉿kali)-[/home/kali]
└─# nmap --top-ports 10  192.168.100.58

Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 07:26 -0400
Nmap scan report for 192.168.100.58
Host is up.

PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
25/tcp   filtered smtp
80/tcp   filtered http
110/tcp  filtered pop3
139/tcp  filtered netbios-ssn
443/tcp  filtered https
445/tcp  filtered microsoft-ds
3389/tcp filtered ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 3.66 seconds
```

## Perform a Sequential Ports Scan

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -r 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 07:29 -0400
Stats: 0:00:27 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 13.00% done; ETC: 07:33 (0:03:01 remaining)
Stats: 0:01:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.00% done; ETC: 07:33 (0:02:20 remaining)
Stats: 0:01:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.50% done; ETC: 07:33 (0:02:19 remaining)
Stats: 0:01:05 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 32.00% done; ETC: 07:33 (0:02:18 remaining)
Stats: 0:01:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 32.50% done; ETC: 07:33 (0:02:17 remaining)
Stats: 0:01:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 33.00% done; ETC: 07:33 (0:02:16 remaining)
```

## Foundational Scanning

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PN 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 07:35 -0400
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.00% done; ETC: 07:38 (0:03:24 remaining)
Nmap scan report for 192.168.100.58
Host is up.
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 201.88 seconds
```

## Ping Only Scan

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -sP 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:17 -0400
Nmap scan report for 192.168.100.58
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.51 seconds
```

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -sP 192.168.100.58/24
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:18 -0400
Nmap scan report for 192.168.100.1
Host is up (0.0059s latency).
MAC Address: 30:E9:8E:DA:3F:50 (Huawei Technologies)
Nmap scan report for 192.168.100.6
Host is up (0.10s latency).
MAC Address: F0:C8:50:0B:46:5F (Huawei Technologies)
Nmap scan report for 192.168.100.15
Host is up (0.00073s latency).
MAC Address: 00:24:D6:E5:29:AF (Intel Corporate)
Nmap scan report for 192.168.100.41
Host is up (0.10s latency).
MAC Address: B6:5E:AF:E4:8B:8B (Unknown)
Nmap scan report for 192.168.100.42
Host is up (0.092s latency).
MAC Address: 90:94:97:2D:A4:03 (Huawei Technologies)
Nmap scan report for 192.168.100.58
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 6.07 seconds
```

## Tcp Syn Scan

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PS 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:21 -0400
Nmap scan report for 192.168.100.58
Host is up.
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 201.88 seconds
```

## Sctp Init Ping

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PE 192.168.100.58
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:36 -0400
Stats: 0:01:49 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 54.00% done; ETC: 11:39 (0:01:34 remaining)
Stats: 0:02:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.50% done; ETC: 11:39 (0:01:22 remaining)
Nmap scan report for 192.168.100.58
Host is up.
All 1000 scanned ports on 192.168.100.58 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 201.90 seconds
```

## Icmp Timestamp Ping

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PP tesla.com
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:51 -0400
Nmap scan report for tesla.com (23.7.244.207)
Host is up (0.31s latency).
Other addresses for tesla.com (not scanned): 2.18.50.207 2.18.54.207 23.40.100.207 2.18.48.207 2.18.52.207 2.18.49.207 2.18.53.207 2.18.55.207 2.18.51.207
rDNS record for 23.7.244.207: a23-7-244-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 23.05 seconds
```

## Icmp Address Mask Ping

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PN tesla.com
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 11:58 -0400
Nmap scan report for tesla.com (23.7.244.207)
Host is up (0.38s latency).
Other addresses for tesla.com (not scanned): 2.18.50.207 2.18.54.207 2.18.49.207 2.18.48.207 23.40.100.207 2.18.52.207 2.18.53.207 2.18.51.207 2.18.55.207
rDNS record for 23.7.244.207: a23-7-244-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 23.28 seconds
```

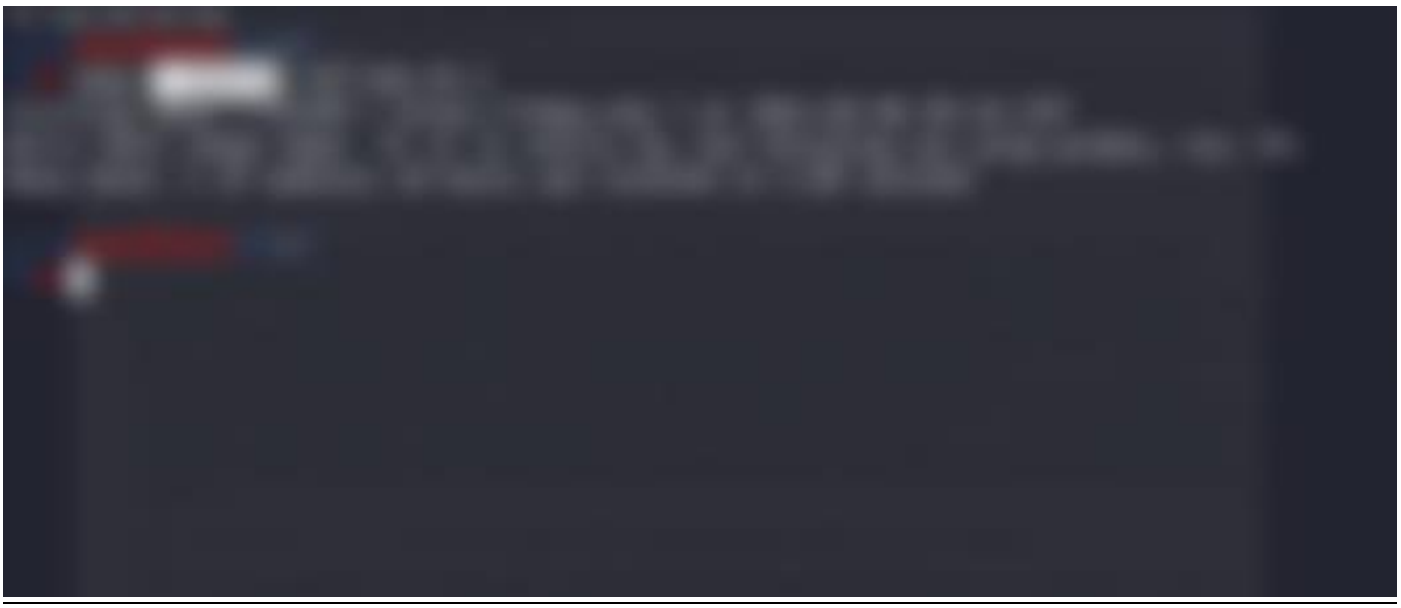
## Ip Protocol Ping

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PO tesla.com
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 12:02 -0400
Nmap scan report for tesla.com (23.40.100.207)
Host is up (0.36s latency).
Other addresses for tesla.com (not scanned): 2.18.53.207 2.18.49.207 2.18.54.207 2.18.50.207 2.18.48.207 2.18.52.207 23.7.244.207 2.18.51.207 2.18.55.207
rDNS record for 23.40.100.207: a23-40-100-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 24.21 seconds
```

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -PR tesla.com
Starting Nmap 7.97SVN ( https://nmap.org ) at 2025-07-23 12:04 -0400
Nmap scan report for tesla.com (2.18.52.207)
Host is up (0.51s latency).
Other addresses for tesla.com (not scanned): 2.18.54.207 23.7.244.207 2.18.53.207 2.18.49.207 23.40.100.207 2.18.50.207 2.18.51.207 2.18.48.207 2.18.55.207
rDNS record for 2.18.52.207: a2-18-52-207.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 25.77 seconds
```

## Advanced Scanning

## Tcp Fine Scan





info@emrchains.com

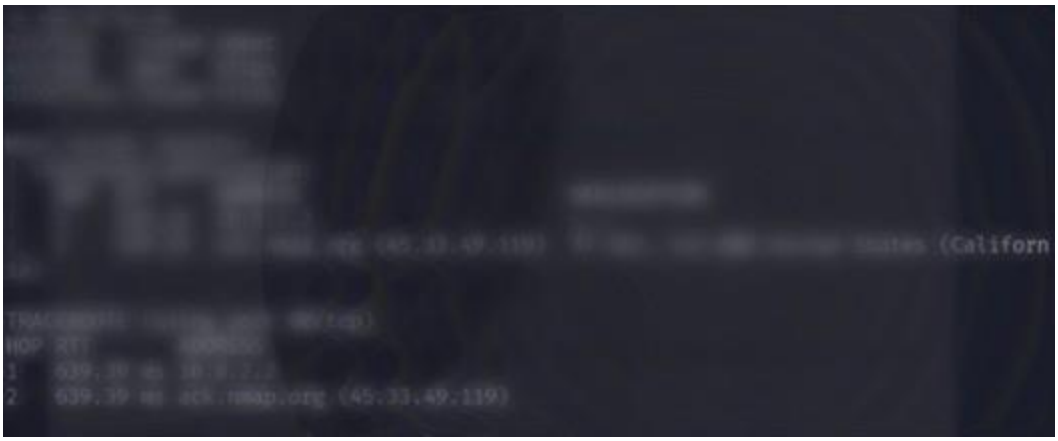
office#1401, wing#04, NSTP, NUST,
H.12, Islamabad, Pakistan

## Nmap Script Engine

info@emrchains.com

office#1401, wing#04, NSTP, NUST,
H.12, Islamabad, Pakistan

SYN Stealth Scan
remaining)
2600:3c01:e000:2e6::6d4e:7061
(no-response)



## 3.Learning Outcome

 **Understanding Nmap Basics**
Learned how Nmap is used for network discovery and security auditing.

 **Types of Scans**
Understood different types of scans such as:

- TCP Connect Scan (-sT)
- SYN Scan (-sS)
- UDP Scan (-sU)
- Service Version Detection (-sV)
- OS Detection (-O)

 **Target Specification**
Learned how to scan specific IPs, IP ranges, and domain names.

 **Scan Results Interpretation**
Gained the ability to read and analyze scan results like open/closed ports and service information.

 **Real-World Use Cases**
Understood how ethical hackers and security professionals use Nmap in penetration testing and vulnerability assessments

## 4.Conclusion

By watching the Nmap video, I gained practical knowledge of how Nmap works and how it can be used to perform powerful network scans. I learned the importance of scanning techniques, how to identify live hosts, open ports, and running services. Overall, this tool is essential for cybersecurity professionals to assess the security posture of networks.